

# TASNIM JAHAN

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[Website](#) — [GitHub](#) — [Google Scholar](#) — [LinkedIn](#)

## Research Interests

• Computer Vision • Medical Image Analysis • Domain Adaptation • 2D/3D Image Segmentation • Representation Learning • Few-Shot Learning • Deep Learning

## Education

**M.Sc. in Computer Science and Engineering (Data Science), CGPA: 3.92 / 4.00** Oct 2021 – Jun 2024  
United International University Dhaka, Bangladesh  
*Summa Cum Laude (Highest Distinction), CGPA: 4.00 / 4.00 in last 5 semesters*

**M.Sc. in Mathematics, CGPA: 3.77 / 4.00** Jan 2016 – Apr 2017  
Comilla University Cumilla, Bangladesh  
*Thesis: Bernoulli's Equation with the Variation of Velocity, Density, and Diameter of Pipe; Advisor: Zillur Rahman*

**B.Sc. in Mathematics, CGPA: 3.61 / 4.00** Mar 2010 – Jan 2016  
Comilla University Cumilla, Bangladesh  
*Project: Numerical Study for Solving Initial Value Problems in Ordinary Differential Equations Using Single-Step and Multi-Step Methods; Advisor: Prof. Dr. Khalifa Mohammad Helal*

**Higher Secondary Certificate (HSC), Science, GPA: 5.00 / 5.00** Jul 2006 – Sep 2008  
Mvi. Shamsul Karim College Feni, Bangladesh

**Secondary School Certificate (SSC), Science, GPA: 5.00 / 5.00** Jan 2004 – Jun 2006  
Chhagalnaiya Pilot Girls' High School Feni, Bangladesh

## Research Experience

**Independent Researcher (Computer Vision & Deep Learning)** Jan 2024 – Present  
Remote

*Advisors: Prof. Andrew King (King's College London, UK); Prof. Swakkhar Shatabda (BRAC University, Bangladesh)*

- Conducting research on few-shot learning, transfer learning, visual representation learning, and cross-domain adaptation using CNNs, Vision Transformers, and pretrained vision foundation models for robust visual learning.
- First author of a Q1 journal manuscript under revision on a lightweight attention-enhanced framework for ultrasound image segmentation.
- Co-author of an IEEE SMC 2025 paper on efficient deepfake detection using sparse subnetworks.

## Publications

- **Tasnim Jahan**, Andrew P. King, Swakkhar Shatabda. [HyProDINO: A Hybrid Prototype Guided DINO Framework for Cross-Domain Few-Shot Medical Image Segmentation](#). Accepted for oral presentation at the MLMI Workshop, MICCAI 2026.
- **Tasnim Jahan**, Md Easin Arafat, Swakkhar Shatabda. ThyCLIPNet: A BiomedCLIP-Guided Lightweight Attention-Enhanced DeepLabV3+ Framework for Robust Thyroid Nodule Segmentation. Under revision (Q1 journal).
- L. Al Amin, Md. Ismail Hossain, Thanh Thi Nguyen, **Tasnim Jahan**, Mahbubul Islam, Faisal Quader. [Uncovering Critical Features for Deepfake Detection through the Lottery Ticket Hypothesis](#). *IEEE SMC*, 2025.

## Awards & Scholarships

**MICCAI 2026 RISE Travel Grant** — Recipient of competitive travel support to attend MICCAI 2026 Aug 2026

**Merit-based Scholarship (5 times)** at United International University for academic performance Jan 2023 – Aug 2024

**Ranked 5th** in LISA 2025 Challenge, RISE-MICCAI Summer School May 2025 – Oct 2025

**Mentee** in RISE-MICCAI Mentorship Program Oct 2025 – Oct 2026

## Projects

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**3D Hippocampus Segmentation (Ultra-low-field MRI)** | *Python, MONAI, PyTorch* | [Code](#) Jul 2025 – Sep 2025

- Developed a MONAI-based 3D deep learning pipeline for bilateral hippocampus segmentation from ultra-low-field MRI.
- Implemented and compared Attention U-Net, DynUNet, SegResNet, and UNETR using class-balanced cropping and sliding-window inference.
- Among the tested models, Attention U-Net achieved the highest validation Dice ( $\approx 0.60$ ) with post-processing.

**Thyroid Nodule Segmentation** | *Python, TensorFlow/Keras* | [Code](#) — [Report](#) Nov 2023 – Feb 2024

- Developed an end-to-end deep learning pipeline for thyroid nodule segmentation on the TN3K ultrasound dataset.
- Implemented and compared U-Net variants with pretrained encoders (VGG16, VGG19, ResNet50, MobileNetV2).
- Best among the tested encoders: MobileNetV2-based U-Net (Dice: 84.70%, IoU: 73.80%), achieving strong accuracy with a lightweight backbone.

**Spatio-Temporal Data Analysis Using Deep Learning (Survey Paper)** | *Literature Review* | [Report](#) Jul 2023 – Nov 2023

- Conducted a structured survey of deep learning methods for spatio-temporal data: data types, pipelines, and models.
- Covered CNNs, RNNs, GNNs, and Transformers for prediction, classification, clustering, and anomaly detection.
- Summarized real-world applications, open challenges, and research directions.

**Hypo-thyroid Disease Prediction** | *Python, Scikit-learn* | [Code](#) — [Report](#) Feb 2022 – May 2022

- Developed a machine learning pipeline for early detection of hypothyroid disorder using clinical and demographic data.
- Applied preprocessing techniques including imputation, feature encoding, scaling, and SMOTE for class imbalance.
- Trained and compared Decision Tree, Logistic Regression, SVM, and Random Forest classifiers.
- Achieved the best performance with Random Forest (Accuracy: 98.70%), outperforming other models.

**Face Mask Detection** | *Python, TensorFlow, OpenCV* | [Code](#) — [Report](#) Mar 2023 – Jul 2023

- Implemented a full pipeline for mask vs. no-mask detection using a MobileNetV2-based CNN.
- Included training, evaluation, and real-time inference with OpenCV.

**Predict Customer's Future Purchases** | *Python, Machine Learning* | [Code](#) — [Report](#) Jun 2022 – Sep 2022

- Developed a machine learning pipeline to predict customer purchases within 90 days using transactional retail data.
- Trained and evaluated Logistic Regression, Decision Tree, Random Forest, and Gaussian Naïve Bayes.
- Achieved the best performance with Logistic Regression (Accuracy: 99.10%, F1-score: 98.50%), while GaussianNB provided the fastest inference.

**Bangla Financial Sentiment Classification** | *Python, ML* | [Code](#) — [Report](#) Jan 2023 – Aug 2023

- Collected and labelled 1.8K+ Bangla financial headlines from the Prothom Alo newspaper for sentiment classification.
- Achieved the best performance with SVM (F1: 78%) and Logistic Regression (Accuracy: 76.60%).

**LLM-Enhanced Stock Movement Prediction** | *ML* | [Report](#) Aug 2023 – Sep 2023

- Integrated financial time-series data with information extracted from news using LLMs for weekly stock prediction.
- Improved Logistic Regression (Accuracy: 54.80%  $\rightarrow$  63%, F1: 42.10%  $\rightarrow$  60.90%).

**IoT-Based Watering Automation and Security System for Plants & Pets** | *Sensors* | [Report](#) Jul 2022 – Nov 2022

- Designed an IoT concept for automated watering with basic security monitoring for plants and pets.
- Focused on automation logic, sensor-driven triggers, and system-level design.

## Skills

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**Programming:** Python, SQL, Java, C, C++

**Deep Learning:** PyTorch, TensorFlow, Keras, MONAI

**Computer Vision / ML:** OpenCV, scikit-learn, NumPy, Pandas

**Research Tools:** LaTeX, Kaggle, Google Colab, Jupyter Notebook

**Visualization:** Matplotlib, Power BI

**Soft Skills:** Complex problem solving, rapid skill acquisition, effective communication

## Relevant Coursework

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- Advanced Artificial Intelligence
- Deep Learning
- Machine Learning
- Data Mining
- Natural Language Processing
- Research Methodology
- Advanced Database Systems
- Differential & Integral Equations
- Discrete Mathematics
- Linear Algebra
- Numerical Analysis
- Ordinary & Partial Differential Equations
- Methods of Statistics
- Vector Calculus & Tensor Analysis

## Teaching Experience

### Section Leader (Volunteer)

Apr 2026 – May 2026

*Stanford University – Code in Place*

*Remote*

- Selected through a competitive process from 2,000 applicants to join a global teaching team.
- Taught Python programming to a diverse cohort in a large-scale online course.
- Led problem-solving sessions and mentored students to improve programming and analytical skills.

### Instructor – Mathematics

Jan 2017 – Mar 2018

*Institute of Computer Science and Technology (ICST)*

*Feni, Bangladesh*

- Taught mathematics to 200+ diploma-level students across engineering disciplines.
- Designed lectures, assignments, and assessments tailored to diverse learning needs.

### Junior Teacher – Mathematics

Mar 2018 – Dec 2018

*Mainamati English School and College*

*Cumilla, Bangladesh*

- Delivered mathematics instruction in English medium and monitored student performance.
- Founded and led a math club to improve student engagement.

## Professional Experience

### Freelance Data Analyst

Jan 2023 – Feb 2024

*Fiverr*

*Remote*

- Developed interactive dashboards and data-driven reports using Microsoft Power BI.
- Performed data cleaning, statistical analysis, and feature engineering on real-world datasets.

### Field Data Collector

Nov 2021 – Dec 2021

*BRAC James P Grant School of Public Health*

*Dhaka, Bangladesh*

- Collected qualitative and quantitative data from ~100 participants using Kobo Toolbox.
- Ensured data quality, consistency, and proper documentation for analysis.

### Programme Organizer – Education

Jan 2022 – Apr 2022

*Jagorani Chakra Foundation*

*Cox's Bazar, Bangladesh*

- Supported the implementation of a UNICEF-funded education program by training teachers in effective teaching methodologies and classroom practices.
- Collaborated with stakeholders, monitored field activities, and contributed to reporting and documentation.

### Education Facilitator – Education

Jan 2019 – Apr 2020

*DanChurch Aid*

*Cox's Bazar, Bangladesh*

- Delivered literacy and numeracy sessions to ~300 learners in a UNICEF-funded program.
- Designed course materials and conducted baseline and endline data collection.

## Academic Service

- **Journal Reviewer**, Biomedical Signal Processing and Control (Q1) — reviewed 6 manuscripts.
- **Workshop Reviewer**, MICCAI 2026 MSB EMERGE and FAIMI-BRIDGE-EPIMI — reviewed 3 manuscripts.

## Online Courses & Certifications

**WorldQuant University:** Computer Vision Lab

Dec 2025 – Present

**RISE-MICCAI Summer School** (MICCAI Society)

Jul 2025

**Stanford University:** Code in Place

Apr 2023 – Jun 2023

**Google:** Data Analytics Specialization

Apr 2021 – Jun 2021

**University of Amsterdam:** Basic Statistics

Jun 2021 – Jul 2021

## Extracurricular Activities

**English & Bangla News Presenter** at Bangladesh Betar, Cox's Bazar, Bangladesh

Apr 2025 – Present

**Member** at Partibartan, Cultural Organization, Comilla University, Cumilla, Bangladesh

Jan 2014 – Jan 2017